

Review of Key Concepts:

Probability:

- $IP(X \leq t) = IP(a \cdot X + b \leq a \cdot t + b)$ for $a > 0, b \in \mathbb{R}$
 $= IP(a \cdot X + b \geq a \cdot t + b)$ if $a < 0$
- CDF's: $F_X(x) = IP(X \leq x)$
- Countable Additivity: For $t > 0$, $IP(|X| \geq t) = IP(\{X \leq -t\} \cup \{X \geq t\})$
 $= IP(X \leq -t) + IP(X \geq t)$

Expected Value & Variance:

- $E[a \cdot X + b] = a \cdot E[X] + b$ for any $a, b \in \mathbb{R}$, r.v. X
- $Var(a \cdot X + b) = a^2 \cdot Var(X)$ for any $a \in \mathbb{R}$, r.v. X
- $E[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = a_1 E[X_1] + \dots + a_n E[X_n]$ for any joint r.v.'s $X_1, \dots, X_n$
- If $X_1, \dots, X_n$ are mutually independent r.v.'s, then
 - + $Var(a_1 X_1 + \dots + a_n X_n) = a_1^2 Var(X_1) + \dots + a_n^2 Var(X_n)$
 - + $E[X_1 \cdot X_2 \cdot \dots \cdot X_n] = E[X_1] \cdot E[X_2] \cdot \dots \cdot E[X_n]$

Random Variables:

- If $X_1, \dots, X_n$ are independent r.v.'s, with marginals of $p_{X_i}(\cdot) / f_{X_i}(\cdot)$, then their joint pmf/pdf is:

$$p(x_1, \dots, x_n) = p_{X_1}(x_1) \cdot p_{X_2}(x_2) \cdot \dots \cdot p_{X_n}(x_n)$$

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) \cdot \dots \cdot f_{X_n}(x_n)$$

Common RV's:

- If $X_1, \dots, X_n$ are iid Bernoulli(α), then $\sum_{i=1}^n X_i$ has a Binomial(n, α) distribution.
- If $X \sim \text{Normal}(\mu, \sigma^2)$, then $a \cdot X + b \sim \text{Normal}(a \cdot \mu + b, a^2 \cdot \sigma^2)$
 - + $\frac{X - \mu}{\sigma} \sim \text{Norm}(0, 1)$
 - + $IP(X \leq t) = IP\left(\frac{X - \mu}{\sigma} \leq \frac{t - \mu}{\sigma}\right) = \Phi\left(\frac{t - \mu}{\sigma}\right)$
- If $X \sim \text{Norm}(\mu_x, \sigma_x^2)$ and $Y \sim \text{Norm}(\mu_y, \sigma_y^2)$ are independent, then
$$aX + bY \sim \text{Norm}(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2) \text{ for } a, b \in \mathbb{R}$$
 - + Same holds for many mutually independent Normal R.V.'s

Sampling:

Imagine we have a population (e.g. cars in NJ), and some numerical quantity of interest (e.g. 1 if electric, 0 o/w).

This quantity has some distribution in the population, in this example, say 5% of cars are electric.

→ This distribution tells us about the frequency with which items (cars) with some property (electric) occur in this population.

If we were to:

random! $\left\{ \begin{array}{l} \text{i) choose an item (car) at random from the population, such that} \\ \text{"}\omega \in S\text{"} \quad \text{all items are equally likely to be chosen.} \end{array} \right.$

not random $\left\{ \begin{array}{l} \text{ii) observe this quantity of interest (1 if electric, 0 o/w) for the} \\ \text{"}X(\omega)\text{"} \quad \text{chosen item (car), and call this } X \end{array} \right.$

→ X is a random variable, whose pmf (in this example) is exactly the distribution of our quantity of interest in the population.

"sampling"

Takeaway: We can model the random result of selecting and observing an item from a population by a r.v. whose pmf/pdf is given by the population's distribution of some quantity.

Different distribution of
a quantity in the population

↔

Different pmf / pdf of the r.v.
used to model the sampling process

e.g. 5% electric
vs
1% electric

↔

$X \sim \text{Ber}(\underline{0.05})$
vs
 $X \sim \text{Ber}(\underline{0.01})$

Key: The parameter of X 's distribution tells us something about the population!

Usually we sample more than one item from the population.

We assume that items are all equally likely to be selected.

We also assume either:

- i) sampling occurs with replacement
- ii) sampling is done without replacement, but from a population that is infinitely large.

Let X_1 = quantity for randomly selected 1st item

X_2 = quantity for randomly selected 2nd item

$\vdots$

X_n = quantity for randomly selected n -th item

} A random sample of size n .

Because of the assumptions in ~~Ⓢ~~ above, notice that

- the pmfs/pdfs of each $X_1, X_2, \dots, X_n$ are exactly the same!
- $X_1, X_2, \dots, X_n$ are mutually independent

→ So, the joint distribution of $X_1, \dots, X_n$ is:

• joint pmf $p(x_1, \dots, x_n) = \underline{p}_X(x_1) \cdot \underline{p}_X(x_2) \cdot \dots \cdot \underline{p}_X(x_n)$, if $X_1, \dots, X_n$ are discrete

• joint pdf $f(x_1, \dots, x_n) = \underline{f}_X(x_1) \cdot \underline{f}_X(x_2) \cdot \dots \cdot \underline{f}_X(x_n)$, if $X_1, \dots, X_n$ are cts.

↑ ↑ ↑
The same marginal pmf/pdf's.

This pmf/pdf is the population distribution.

Def. (Independent and Identically Distributed): We say that $X_1, \dots, X_n$ are iid (indep. and identically distributed) according to pmf $p_X(\cdot)$ (pdf $f_X(\cdot)$) if their joint pmf (pdf) equals:

$$p(x_1, \dots, x_n) = \prod_{i=1}^n p_X(x_i) \quad \left(f(x_1, \dots, x_n) = \prod_{i=1}^n f_X(x_i) \right)$$

for all $x_1, \dots, x_n$.

Note: If $p_X(\cdot)$ is the pmf of a $\text{Ber}(\alpha)$ random variable, then we would write " $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\alpha)$ " to denote that $X_1, \dots, X_n$ are i.i.d. with each having a (marginal) $\text{Bernoulli}(\alpha)$ distribution.

Likewise, we may write " $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ " for n independent $\text{Exp}(\lambda)$ random variables.

Note: Having iid random variables is an assumption, never something we will be checking. However, it will be assumed near-constantly in the rest of the course.

Takeaway: When we say " $X_1, \dots, X_n$ are iid", think of sampling n items from a population (assuming $\odot$), where the distribution of each $X_1, \dots, X_n$ is the population distribution.

Before we obtain any data (i.e. before choosing items), there is uncertainty.

$\leadsto$ Represented by the random variables $X_1, \dots, X_n$.

But after we choose the n items from the population, there is no uncertainty!

$\leadsto$ Represented by the realizations $x_1, \dots, x_n$ (of r.v.'s $X_1, \dots, X_n$)

Note: The r.v.'s $X_1, \dots, X_n$ are often called a "random sample"

The realizations $x_1, \dots, x_n$ are often called just a "sample"

Samples and Statistics:

Consider the population of current NJ-registered cars.

We choose, assuming $\textcircled{*}$, a random sample of 4 cars from this population, and their m.p.g. will be denoted X_1, X_2, X_3, X_4 .

Say that I do this, and obtain the following sample

$$x_1 = 33.2 \quad x_2 = 22.7 \quad x_3 = 26.5 \quad x_4 = 30.9$$

Any quantity we can compute from sample data is called a statistic. Some examples:

- Sample mean $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$
- Sample range $(\max_i x_i) - (\min_i x_i)$

After data is obtained, the value of a statistic, e.g. the sample mean, is not random! In our case, $\bar{x} = 28.325$.

But prior to obtaining data, there is uncertainty as to what the value of a statistic will be.

Therefore, a statistic is a random variable!

I.e.: Different choice for car #1



(Possibly) Different realization
 x_i^* of the r.v. X_i



(Possibly) Different value for
the sample mean $\bar{x}$

The distribution of a statistic is called its sampling distribution. It can depend on:

- The population distribution || The most important!!
- The size of a sample
- The method of sampling || This won't be a concern for us

Much of statistics (as a field) is concerned with finding the sampling distribution of different statistics.

→ Usually, this is very difficult. But some settings are special!

Let: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ denote the random variable constructed by averaging the random sample $X_1, \dots, X_n$.

If $X_1, \dots, X_n$ are iid, then:

- $E[\bar{X}_n] = E[X_1] (= E[X_2] = \dots = E[X_n])$
- $Var(\bar{X}_n) = \frac{1}{n} \cdot Var(X_1) (= \frac{1}{n} Var(X_2) = \dots = \frac{1}{n} Var(X_n))$

Proof: Done in HW.

Recall: Variance is a measure of how "spread out" a random variable can be.

If a r.v. has zero variance, then it is a constant!

Informal Result: "As n gets large, $\bar{X}_n$ looks more like a constant (i.e. is less and less random)"

Thm. (Law of Large Numbers, LLN): If $X_1, \dots, X_n$ are iid random variables, where $E[X_i] = \mu$ ($= E[X_1] = \dots = E[X_n]$ since r.v.'s are iid),

then:

$$\bar{X}_n \xrightarrow{n \rightarrow \infty} \mu.$$

Informal Reasoning:

- $E[\frac{1}{n} \sum_{i=1}^n X_i] = \mu$ for all n
- $\text{Var}(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{\text{Var}(X_1)}{n}$, converges to 0 as $n \rightarrow \infty$.

Observation #1: The quantity $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is random, so "convergence" here is a bit more nuanced. Not a concern for us, nor something we will discuss further. Instead, just something to be aware of: $\bar{X}_1, \bar{X}_2, \dots$ is a sequence of r.v.'s, not real numbers.

Obs. #2: The only condition required here was that $X_1, \dots, X_n$ are iid random variables. This will differ from later results!

Corollary: If $X_1, \dots, X_n$ are iid random variables and $f(\cdot)$ is any real-valued function, then:

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \xrightarrow{n \rightarrow \infty} E[f(X_1)]$$

Note: This is still random!

$= E[f(X_1)] = \dots = E[f(X_n)]$ since r.v.'s are iid.

So far, we know several things about the random sample mean $\bar{X}_n$ of n iid random variables. Let $\mu = E[X_1]$, $\sigma^2 = \text{Var}(X_1)$

- $E[\bar{X}_n] = \mu$ (shown in HW4)
- $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$ (shown in HW4)
- $\bar{X}_n \xrightarrow{n \rightarrow \infty} \mu$ (LLN)

Changing the "population distribution" can change μ, σ^2 , hence changing the distribution of $\bar{X}_n$

[Changing the sample size n also changes $\text{Var}(\bar{X}_n)$]

Can we say anything more? Let's look at a special case first.

Ex.: Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, for some $\mu \in \mathbb{R}$, $\sigma^2 > 0$.

With $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, we know (from before) that

$$\bullet \mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

But $X_1, X_2, \dots, X_n$ are (mutually) independent Normal R.V.'s.

As a special property of Normal R.V.'s, their linear combination is also a Normally distributed r.v.!

In General:

If $X_1 \sim N(\mu_1, \sigma_1^2)$, $X_2 \sim N(\mu_2, \sigma_2^2)$, ..., $X_n \sim N(\mu_n, \sigma_n^2)$ are mutually indep. r.v.'s, then:

$$\sum_{i=1}^n X_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

[This is the key - recall that $\mathbb{E}[\sum_{i=1}^n X_i] = \sum_{i=1}^n \mathbb{E}[X_i]$ always.

Takeaway:

If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, then $\bar{X}_n \sim N(\mu, \frac{\sigma^2}{n})$

What if $X_1, \dots, X_n$ are still iid, but not normally distributed?

Thm. (Central Limit Theorem, CLT): Let $X_1, \dots, X_n$ be iid, and $\mu = \mathbb{E}[X_i]$, $\sigma^2 = \text{Var}(X_i)$. If $\sigma^2 < \infty$, then:

• $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$ has an approximate $N(0,1)$ distribution.

• As n grows, the approximation becomes better.